Do MOL Bubi trips connect areas to activities that are scarce locally?

Problem and gap

Jacobs (1961) emphasises the importance of mixed urban uses, bringing homes, workplaces, shops and other activities into close proximity. Such a mix can support urban economic and social life, for example, by sustaining local businesses and creating opportunities for people to meet and interact. However, not every neighbourhood contains the same range of activities. Where particular activities are scarce locally, transport connections can extend the opportunities available to people beyond their immediate neighbourhood. Geurs and van Wee (2004) provide a framework for understanding this through accessibility, linking the spatial distribution of activities with the transport systems that enable people to reach them. The proposed research would investigate one observable part of this phenomenon using trip records from Budapest's MOL Bubi bike-sharing system. It would examine whether MOL Bubi trips disproportionately connect areas to activities that are relatively scarce at their origins.

Previous studies have examined the relationship between cycling and urban activities in other cities. Zhang et al. (2018), for example, combine bicycle rental records and points of interest to identify urban functional zones. Chen et al. (2022) take a network approach, representing areas as nodes and bike-sharing trips as links, then using community detection to identify strongly connected activity zones. This proposal draws on both approaches: it uses independently described activity profiles and a network of observed trips to examine whether areas connect to activities that are relatively scarce locally.

Recent Bubi research provides a local starting point. Korompay, Földes and Csiszár (2026) analyse station-level arrival and departure patterns and relate them to surrounding land uses. The proposed study would retain origin-destination connections as a network, rather than aggregating trips into station profiles. It would therefore examine not only how each area is used, but which areas it connects to and whether those destinations offer functions that are relatively scarce at the origin.

These studies provide a foundation for understanding urban functions and bike-sharing networks. However, they do not directly address whether MOL Bubi connects areas to activities that are scarce locally. The proposed study addresses this question by examining the activity profiles at both ends of each connection, while accounting for distance and overall trip volumes.

Research question and hypothesis

Do MOL Bubi trips tend to connect areas to destinations where activities that are scarce at the origin are more available?

The hypothesis is that areas lacking a particular activity have stronger connections to places offering it. For example, areas with few educational facilities may send a larger share of their trips to areas with schools or universities. Looking at the trip network shows not just how many trips each area generates, but where those trips go and what activities are available there. The analysis would compare these connections with what would be expected from distance and each area's usual departure and arrival volumes. This would help assess whether the pattern goes beyond people simply travelling to nearby or busy destinations.

Data and proposed approach

The research would combine MOL Bubi origin-destination records with small spatial zones from Budapest's EFM transport model. A group of EFM variables describes travel attraction associated with residential, work, education, shopping, healthcare, leisure and other activities. Their units and comparability across categories would need to be assessed. If unsuitable, OpenStreetMap points of interest could provide an alternative description of each zone's activities.

In the proposed network analysis, EFM zones would serve as nodes, with their activity profiles recorded as node attributes. Observed Bubi trips would form directed links weighted by trip count. The paper by Chen et al. (2022) provides a methodological starting point for working with geographical areas rather than only stations.

For each origin zone, its activity profile would be compared with the trip-weighted profiles of its destinations. A possible network measure is the share of outgoing trips reaching zones where locally scarce activities are more strongly represented. These patterns would be compared with a baseline accounting for distance, destination

activity levels and overall departure and arrival volumes. The exact statistical model would be selected after inspecting the data.

Expected results and practical value

If the hypothesis holds, the network would show connections to activities that are less represented locally, beyond simple geographical proximity and destination popularity. If it does not, that would also be informative: Bubi patterns may be explained more strongly by distance and the concentration of trips around major destinations.

For BKK, the study could add a functional perspective to trip-volume analysis: not only how much the service is used, but which types of areas and activities it connects. Although the results would not establish individual trip purposes, they could help identify locations for closer investigation when planning service improvements or expansion.

References

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